

**2** During the development of a human, the structure of haemoglobin changes.

In the developing fetus the main oxygen-carrying molecule is fetal haemoglobin.

Towards the end of fetal development, fetal haemoglobin is replaced by adult forms of haemoglobin.

- (a) Blood may contain 5.3 million red blood cells in  $1 \text{ mm}^3$  of blood.

One red blood cell can contain 270 million haemoglobin molecules.

Calculate the number of haemoglobin molecules in  $1 \text{ mm}^3$  of blood.

Express your answer in standard form.

(1)

Answer .....

- (b) Which of the following statements about the structure of haemoglobin are correct?

1. It is a fibrous protein
2. It has quaternary structure
3. It contains a non-protein haem group

(1)

- ☐ **A** 1 only
- ☐ **B** 3 only
- ☐ **C** 1 and 2 only
- ☐ **D** 2 and 3 only



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- (ii) The oxygen affinity of fetal haemoglobin is different from adult haemoglobin.

Explain why this difference ensures the fetus obtains enough oxygen during development.

(2)

(Total for Question 2 = 7 marks)

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